Not All Heterogeneity Is Value Conflict: Gradient Interference in Group-Aware RLHF

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Abstract

Standard RLHF trains a single reward model on aggregated preference data, and its weak alignment for minority groups is commonly read as evidence of an inherent fairness-utility trade-off. We argue this framing is incomplete: the failure can arise from gradient conflict during optimization, even when group preferences are compatible. Using a controlled setting that continuously increases the divergence between two group rewards, we find that single-reward RLHF undergoes a sharp worst-group collapse even when the group rewards remain positively correlated. Near the collapse, the aggregated PPO update conflicts with the neglected group's gradient at the optimization level. We then compare two corrections. Max-Min RLHF optimizes the worst-group objective and partially repairs the collapse, but it does not explicitly resolve conflicting components between group gradients. In contrast, a gradient-surgery method (PCGrad-PPO) removes the conflicting gradient components and recovers the worst-group performance more strongly and stably. These results suggest that group-alignment failures often framed as fairness-utility trade-offs are, in an identifiable regime, gradient-interference failures of multi-objective optimization. Addressing the interference directly can outperform worst-group objective selection.

1 Introduction

Reinforcement Learning from Human Feedback (RLHF) has become a standard paradigm for aligning large language models with human preferences (Christiano et al., 2017; Stiennon et al., 2020; Bai et al., 2022; Ouyang et al., 2022). Standard RLHF trains a single reward model on aggregated preference data, and optimizes a policy against the pooled reward. This approach compresses heterogeneous preferences into a single optimization target.

A common interpretation is that heterogeneous preferences create an inherent fairness-utility trade-off: improving alignment for one group must come at the expense of another (Bakker et al., 2022; Liu et al., 2024). Under this view, weak minority alignment arises because the aggregate reward mainly reflects majority preferences.

This framing confuses two distinct notions of conflict. Value-level conflict concerns the reward geometry: whether satisfying one group necessarily hurts another. Optimization-level interference concerns the update geometry: whether the policy update induced by an aggregate objective harms one group even when the group rewards remain compatible. We show that these two notions can be separated in group-aware RLHF. The preferences need not conflict; the RLHF updates can.

We study this mechanism in a controlled setting where preference divergence increases continuously. Using standard RLHF, one group's reward can collapse sharply while the other remains high, even when the two groups' rewards are still positively correlated. Near this collapse, the aggregated PPO update becomes conflicting with the neglected group's gradient. This suggests the failure is optimization-induced gradient interference, not an unavoidable fairness-utility trade-off.

Our contributions are as follows. (i) We construct a controlled benchmark that varies the geometry between two group rewards, separating preference heterogeneity from value-level conflict. (ii) We show that single-reward RLHF can collapse abruptly for the worst group while group rewards remain positively correlated. (iii) We diagnose this collapse at the update level: near the collapse point, the KL-regularized PPO update becomes poorly aligned or even anti-aligned with the neglected group's reward gradient direction. (iv) We compare objective-level and gradient-level corrections. Max-Min RLHF partially repairs the collapse through worst-group objective selection, while PCGrad-PPO repairs it more strongly by directly modifying conflicting update directions.

1.1 Related Work

Pluralistic and Group-aware RLHF. Recent work studies alignment under heterogeneous human preferences. One line of work treats preference aggregation as a collective decision problem and designs group-aware objectives, including pluralistic reward modeling, Max-Min RLHF, and group-robust preference optimization (Chen et al., 2024; Chakraborty et al., 2024; Ramesh et al., 2024; Shen et al., 2025). Another line formulates alignment as multi-objective RLHF, where different preference dimensions are modeled explicitly and combined through linear or nonlinear aggregation (Wang et al., 2024b,a; Xiong and Singh, 2025). A third line pursues personalization by conditioning reward models or policies on user or group specific information (Jang et al., 2023; Poddar et al., 2024; Li et al., 2024; Park et al., 2024). We ask which failure mode these group-aware objectives correct: value-level conflict between preferences or optimization-level interference in the update.

Multi-task Optimization and Gradient Interference. Algorithms for multi-task learning train a single model that can perform well across various tasks (Caruana, 1997; Ruder, 2017; Espeholt et al., 2018; Collobert and Weston, 2008). A key challenge is gradient dominance, where task gradients point in opposing directions, and some tasks dominate others in the aggregated gradients (Kendall et al., 2018). Various methods address this by modifying gradient directions to remove destructive interference (Yu et al., 2020), rescaling gradient magnitudes to balance learning speeds (Suteu and Guo, 2019; Sener and Koltun, 2018; Chen et al., 2018), or finding Pareto optimal update directions that avoid harming any single objective (Lin et al., 2019; Liu et al., 2025). Our setting exhibits analogous dynamics: each preference group defines a distinct optimization objective, and the aggregated reward signal can be dominated by one group.

2 Preliminaries and Experiment Settings

Let $x \in \mathcal{X}$ denote an input prompt and $y \in \mathcal{Y}$ a generated response. A policy $\pi_\theta(y \mid x)$ parameterized by θ defines a conditional distribution over responses. For a given prompt x, annotators prefer response y^+ over y^-.

2.1 Single-Reward RLHF Pipeline

(1) Supervised Fine-Tuning (SFT). Starting from a pretrained model, a reference policy π_{SFT} is obtained by fine-tuning on a dataset of demonstrations using maximum likelihood estimation.

(2) Reward Model Training. A reward model $r_\phi(x, y)$ is a map from prompt-response pairs to scalar scores, trained against the Bradley-Terry style objective (Bradley and Terry, 1952):

$$\mathcal{L}_{\mathrm{RM}}(\phi) = -\log \sigma\big(r_\phi(x, y^+) - r_\phi(x, y^-)\big), \quad (1)$$

where $\sigma(\cdot)$ denotes the sigmoid function.

(3) Policy Optimization. The policy is optimized to maximize the learned reward while remaining close to the reference policy π_{ref}. The objective is:

$$\mathcal{J}_{\mathrm{RLHF}}(\theta) = \mathbb{E}_{\substack{x\sim\mathcal{D} \\ y\sim\pi_\theta}}\big[r_\phi(x, y)\big] - \beta\,\mathrm{KL}(\pi_\theta \| \pi_{\mathrm{ref}})\,, \quad (2)$$

where $\beta > 0$ controls deviation from π_{ref} .

2.2 Max-Min RLHF

In reality, users may have heterogeneous preferences and can be clustered into subpopulations. We assume the population is partitioned into G groups, indexed by $g \in [G] := \{1, \ldots, G\}$. Preference data consist of tuples (x, y^+, y^-, g), and each group has an intrinsic reward function parametrized by ϕ_g. Max-Min RLHF (Chakraborty et al., 2024) optimizes the worst-performing group reward:

$$\mathcal{J}_{\mathrm{MaxMin}}(\theta) = \min_{g\in[G]} \mathbb{E}_{\substack{x\sim\mathcal{D} \\ y\sim\pi_\theta}}\big[r_{\phi_g}(x, y)\big] - \beta\,\mathrm{KL}(\pi_\theta \| \pi_{\mathrm{ref}})\,. \quad (3)$$

Since both group rewards explicitly depend on response length in our experiments, we use the length-invariant mean KL in both single-reward RLHF and MaxMin-RLHF: $\frac{1}{T}\sum_{t=1}^{T} \log \frac{\pi_\theta(a_t|s_t)}{\pi_{\mathrm{ref}}(a_t|s_t)}$. This choice would decouple the regularization term from length and allow human preference to be the sole source of the conciseness signal.

2.3 Projecting Conflicting Gradients

RLHF under heterogeneous preferences can be modeled as a multi-task learning problem, where each annotator group defines a distinct objective. However, jointly optimizing across groups often leads to destructive gradient interference when task gradients conflict. Projecting Conflicting Gradients (PCGrad, (Yu et al., 2020)) addresses this by projecting each gradient with negative cosine similarity onto the normal plane of the other, removing the conflicting component while leaving

non-conflicting gradients unaltered. Given the original task gradient $\mathbf{g}_i = \nabla_\theta \mathcal{J}_i(\theta)$, the projected gradient $\mathbf{g}_i^{\mathrm{PC}}$ is obtained by iteratively removing its components conflicting with the gradients of all other tasks. Specifically, if $\mathbf{g}_i^{\mathrm{PC}} \cdot \mathbf{g}_j < 0$

$$\mathbf{g}_i^{\mathrm{PC}} \leftarrow \mathbf{g}_i^{\mathrm{PC}} - \frac{\mathbf{g}_i^{\mathrm{PC}} \cdot \mathbf{g}_j}{\|\mathbf{g}_j\|^2} \mathbf{g}_j. \tag{4}$$

2.4 Experimental Design

Our experiments build on the IMDb movie review dataset (Maas et al., 2011). Each prompt-response pair (x, y) is associated with two scalar attributes: $r_s(x, y)$, a *positive sentiment alignment* score; and $r_c(x, y)$, a normalized *conciseness* score. Score construction details are provided in Appendix A. We construct two synthetic preference groups: a *majority* group that prioritizes positive sentiment and a *minority* group that prioritizes conciseness. The majority proportion is fixed at 80%, reflecting an imbalanced but realistic population structure. Group-specific reward functions are defined as combinations of the two attributes:

$$r_{\mathrm{maj}}(x, y) = \alpha\, r_s(x, y) + (1 - \alpha)\, r_c(x, y),$$
$$r_{\mathrm{min}}(x, y) = (1 - \alpha)\, r_s(x, y) + \alpha\, r_c(x, y),$$

where $\alpha \in [0.5, 1.3]$ controls the degree of preference divergence between groups. When $\alpha > 1.0$, the two group rewards are in direct conflict, as improving one explicitly penalizes the other. In single-reward RLHF, the reward model is trained on the aggregated dataset. In both MaxMin-RLHF and PCGrad-PPO, a separate reward model is trained for each group and used to train the policy under their respective losses. More detailed implementation are shown in Appendix B.

3 Results and Analysis

We examine how RLHF behaves as group preference divergence increases. Figure 1 compares the main method using worst-group reward. We first show that single-reward RLHF exhibits sharp worst-group collapse even before the group rewards become negatively correlated. We then diagnose this collapse through update-level gradient alignment. Finally, we compare two correction mechanisms: MaxMin-RLHF changes objective selection, while PCGrad-PPO changes the gradient update geometry. We provide additional results comparing single-reward RLHF, MaxMin-RLHF and PCGrad-PPO across different values of the preference divergence parameter α in Appendix C.3.

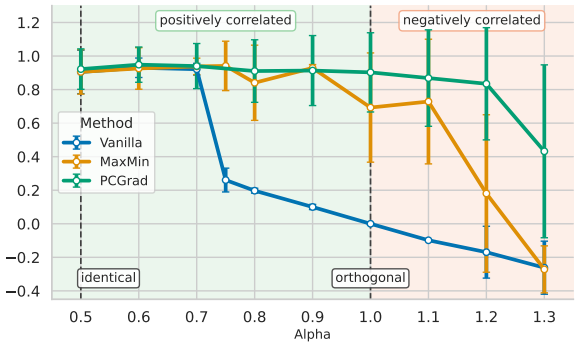


Figure 1: **Worst-group reward under preference heterogeneity.** Vanilla RLHF suffers a sharp worst-group collapse once preference heterogeneity becomes sufficiently large. Max-Min RLHF partially improves worst-group performance, while PCGrad-PPO provides stronger recovery across the heterogeneous regime.

3.1 Single-Reward RLHF Collapses Despite Compatible Rewards

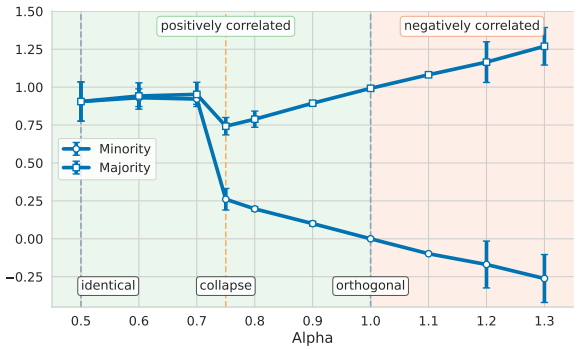


Figure 2: **Vanilla RLHF performance across preference heterogeneity.** Each point represents the mean score, and the vertical bars denote standard deviation. Minority reward collapses sharply once preference divergence becomes sufficiently large.

Figure 2 reports the performance of single-reward (vanilla) RLHF across different values of α. When $\alpha = 0.5$, the two groups have identical preferences, and vanilla RLHF achieves high rewards for both majority and minority groups. However, the failure emerges sharply as preference divergence increases: as α increases from 0.7 to 0.75, the minority reward drops abruptly while the majority reward remains high. Importantly, this collapse occurs at $\alpha = 0.75$, where the empirical Pearson correlation between the two group rewards is still 0.65, inside the positively correlated regime. Thus, the collapse cannot be explained solely by value-level conflict between group rewards. When $\alpha \geq 1.0$, the two groups become orthogonal or conflicting, and vanilla RLHF continues to improve

majority reward while minority reward falls below zero. Overall, this sharp transition suggests that vanilla RLHF undergoes an *optimization-level collapse*: beyond a moderate level of preference heterogeneity, the aggregated PPO update no longer improves the minority objective and instead follows a direction dominated by the majority signal.

Collapse Corresponds to Update-Level Interference To test this mechanism directly, we measure the cosine similarity between the vanilla PPO update direction and the minority group reward gradient. Figure 3 shows that the cosine similarity drops below -0.5 in the early training stages when $\alpha = 0.75$ and $\alpha = 0.8$, indicating that the policy is being updated in a direction that harms the minority group's reward. We demonstrate that this is not affected by the KL term in Appendix C.1. The cosine similarity is most negative in the early training stage, where the sharp performance drop is initiated. It later returns toward positive values, indicating that this diagnostic is most informative about the onset of collapse rather than the final converged policy. This provides evidence of update-level interference: the actual KL-regularized PPO update can move against the neglected group's reward gradient direction near the collapse point.

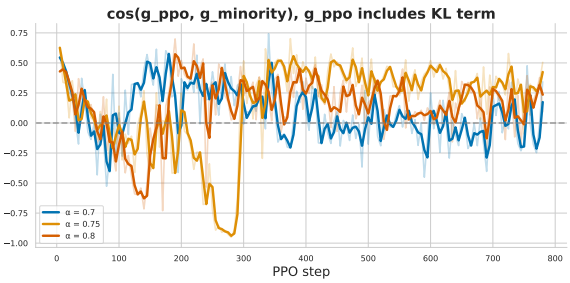


Figure 3: **Gradient alignment between KL-regularized vanilla PPO and minority objective.** The plot reports the cosine similarity between the vanilla PPO update direction, including the KL regularization term, and the minority group reward gradient.

3.2 Objective-Level Selection vs. Gradient-Level Repair

We next compare two ways of correcting the worst-group collapse: selecting a different scalar objective and modifying the group-gradient composition directly. As shown in Figure 1, MaxMin-RLHF improves worst-group reward after the vanilla baseline begins to fail, especially in the intermediate regime where group preferences are divergent but still positively correlated. This shows that the collapse has a clear worst-group consequence and that group-aware optimization is beneficial.

However, MaxMin-RLHF acts through objective selection. It optimizes the currently worst-performing group, and in practice its active objective alternates between the majority and minority groups as their relative performance changes during training. We provide additional evidence of this switching behavior in Appendix C.2. Such switching can improve the worst group locally, but it does not explicitly remove conflicting components between group gradients. Thus, MaxMin-RLHF changes the active scalar objective and therefore changes the update direction indirectly, but it does not directly address the gradient-interference mechanism diagnosed above.

PCGrad-PPO instead keeps the group-specific PPO losses separate during gradient computation and removes components of one group gradient that conflict with another. As shown in Figure 1, it achieves higher worst-group reward than both vanilla RLHF and MaxMin-RLHF across the heterogeneous regime, including cases where group preferences are slightly conflicting. This improvement is notable because PCGrad-PPO does not explicitly optimize the worst-group objective; rather, it keeps both group objectives active while modifying how their gradients are combined. The comparison suggests that the collapse is not only a consequence of using the wrong scalar fairness objective, but also of destructive gradient interference during policy optimization. In this setting, directly mitigating gradient interference is more effective than worst-group objective selection alone.

4 Conclusion

We separate value-level conflict from optimization-level interference in group-aware RLHF. In our controlled benchmark, single-reward RLHF collapses for the worst group even when group rewards remain positively correlated. This collapse is associated with the KL-regularized PPO update becoming anti-aligned with the neglected group's reward-gradient direction. Max-Min RLHF partially repairs the failure through worst-group objective selection, but does not remove conflicting components between group gradients. PCGrad-PPO provides stronger recovery by modifying the group-gradient composition. These findings suggest that group-aware RLHF should be analyzed not only through fairness objectives and aggregation rules, but also through policy-update geometry.

Limitations Our study is controlled and diagnostic rather than a full deployment study. We use synthetic preference groups constructed from sentiment and conciseness attributes on IMDb movie reviews, so the majority and minority groups should not be interpreted as real demographic communities. The experiments are conducted with GPT-2 as model backbone and a two-group setting. Behavior under larger language models, real preference datasets, and multi-group settings need to be explored. The transition between compatible and conflicting preferences may also depend on reward-model quality and PPO hyperparameters used in our setup. Therefore, our results should be interpreted as evidence for a specific optimization-induced failure mode of aggregated RLHF, not as a universal characterization of all heterogeneous preference alignment problems.

Ethical Considerations This work studies group-aware RLHF to understand when aggregate preference optimization may neglect a minority preference. A potential risk is that synthetic preference groups may oversimplify real-world preference diversity or be mistaken for demographic groups. We emphasize that our majority and minority groups are constructed from task attributes and are not intended to represent protected demographic categories. Another risk is that group-aware alignment methods could be misused to target or manipulate specific user groups if deployed without consent or oversight. Our experiments use existing public movie-review data and do not collect new human-subject data or infer personal attributes. The conclusions are intended for diagnostic and methodological understanding rather than direct deployment.

Artifacts and Reproducibility. We use the IMDb movie review dataset, a GPT-2 model fine-tuned, and a DistilBERT sentiment classifier fine-tuned on the IMDb dataset. We cite the creators of these artifacts and use them only for research benchmarking. Where license or intended-use information is available, we follow the corresponding terms; where documentation is incomplete, we restrict our use to controlled experimental analysis and do not redistribute restricted raw data. We report model checkpoints, data construction, hyperparameters, hardware, and evaluation settings in Appendix A.

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Appendix

A Detailed Experiment Settings

Our experiments build on the IMDb movie review dataset (Maas et al., 2011). In this task, Chakraborty et al. (2024) observe that the majority of users prefer sentiment-aligned reviews, while a minority group prefers concise responses. They demonstrate that a single-reward RLHF objective cannot simultaneously align with both criteria, and propose MaxMin-RLHF to optimize the worst-group reward objective, producing responses that better accommodate both subpopulations. Building on these findings, we extend the IMDb benchmark from a fixed heterogeneity setting to a controlled family of heterogeneity regimes.

Following Rafailov et al. (2023), each prompt consists of the first 2–8 tokens of a movie review. For each prompt x, we generate candidate responses using a GPT-2 (Radford et al., 2019) model fine-tuned on IMDb reviews[1]. Each response y is associated with two scalar attributes:

- $r_s(x, y)$: a **positive sentiment alignment** score, obtained from a distilled BERT (Devlin et al., 2019) classifier[2];
- $r_c(x, y)$: a normalized **conciseness** score, computed as a function of response length.

To be specific, $r_c(x, y) = 1 - \frac{l(y)}{l_{\max}}$, where $l(y)$ is the response length and $l_{\max}$ is the maximum response length, set as 512 in this experiment.

Length Bias and Attribution in RLHF. Because one of our group rewards is based on conciseness, response length is a potential confound. Prior work shows that RLHF can exploit response length and that reward-model gains can be partially driven by length correlations rather than semantic quality (Singhal et al., 2024). We therefore report length statistics and use length-normalized KL to separate improvements in the conciseness reward from artifacts introduced by length-sensitive regularization.

Pearson Correlations Curve. To quantify how aligned the two group reward functions are under a given α, we compute their Pearson correlation analytically from the base data distribution. Concretely, we use the raw generation dataset to obtain a pool of (positivity, conciseness) score pairs. For each value of α, we instantiate the majority and minority oracle rewards over this fixed pool, then compute their Pearson correlation (r). Sweeping α from 0.5 to 1.3 yields the curve in Figure 4: at $\alpha = 0.5$ both rewards are identical ($r = 1$); at $\alpha = 1.0$ they reduce to the two raw scores and correlation reduces to that between positiveness and conciseness; and beyond $\alpha = 1.0$ the objectives increasingly conflict, driving the correlation negative. Specifically, we observe that vanilla RLHF collapses near $\alpha = 0.75$, where the Pearson correlation is 0.65.

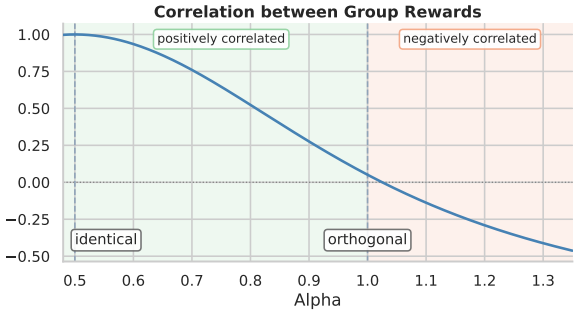


Figure 4: **Pearson Correlations across preference divergence.**

B Implementation Details

To allow for comparability of results, all tools were evaluated on equal-cost hardware, an Ubuntu 24.04.3 LTS server with dual AMD EPYC 9J14 96-core processors, 792 GB RAM, and four NVIDIA GeForce RTX 5090 GPUs with 32 GiB memory each. The NVIDIA driver version is 570.211.01 and the CUDA version is 12.8. Each training or evaluation run was assigned one GPU. The combined training and evaluation run takes approximately one GPU hour, for each α and all methods. CPU usage was not explicitly pinned; the data preprocessing and tokenization code uses up to 32 worker processes, where enabled.

MaxMin-RLHF is implemented from `https://github.com/CharlesQ9/MaxMinRLHF`. The PPO update is implemented using the TRL library (von Werra et al., 2020). Unless otherwise stated, all RLHF experiments use the IMDB dataset with a minority population of 20% and a majority population of 80%. Policies are initialized from GPT-2, with the reference policy and value model initialized from the same backbone. We train for 50,000 PPO episodes with 3 PPO epochs per batch, an

[1] https://huggingface.co/lvwerra/gpt2-imdb
[2] https://huggingface.co/lvwerra/distilbert-imdb

effective batch size of 64, bf16 precision, reward whitening, cosine learning-rate decay, and learning rate 3×10^{-6}. The KL coefficient is fixed to $\beta = 0.1$ using mean per-token KL. Generation during training uses a maximum of 512 new tokens, and evaluation uses 256 new tokens.

MaxMin-RLHF Implementation Details. We implement MaxMin-RLHF by training a separate reward model for each group. During PPO, each rollout batch B is scored by both group reward models. Let $r_g(x_i, y_i)$ denote the reward assigned by group g to response y_i for prompt x_i. We compute the batch-level mean reward for each group,

$$\bar{r}_g = \frac{1}{|B|} \sum_{(x_i,y_i)\in B} r_g(x_i, y_i),$$

and select the worst performance group

$$g^\star = \arg\min_g \bar{r}_g.$$

The PPO update then uses $r_{g^\star}(x_i, y_i)$ as the scalar reward for every sample in the batch. Thus, MaxMin-RLHF performs batch-level worst-group selection rather than taking a per-sample minimum across groups. In practice, we evaluate both group reward heads on the same generated responses, extract the reward before the final non-padding token, and optimize the selected group's rewards with the standard PPO objective and KL regularization.

PCGrad-PPO Implementation Details. PCGrad-PPO treats the two group rewards as separate policy gradient objectives instead of reducing them to a single scalar objective. For each PPO rollout batch, we compute both group rewards r_1 and r_2. We then construct separate advantage streams for the minority group reward, the majority group reward, and the KL regularization term. The clipped PPO policy losses from the two group reward advantage streams are used to compute the group gradients to which PCGrad is applied.

Let g_1 and g_2 be the policy gradients obtained from the two reward losses. If the gradients conflict, i.e.,

$$g_1^\top g_2 < 0,$$

we apply PCGrad projection:

$$g_1^{\text{PC}} \leftarrow g_1 - \frac{g_1^\top g_2}{\|g_2\|^2} g_2, \qquad g_2^{\text{PC}} \leftarrow g_2 - \frac{g_1^\top g_2}{\|g_1\|^2} g_1.$$

If $g_1^\top g_2 \geq 0$, both gradients are left unchanged. The KL regularization is then computed. The final policy gradient is the sum of the projected group gradients plus the KL regularization. The value function is updated with the standard PPO value loss and is not modified by PCGrad.

C Additional Results.

C.1 Additional Cosine Similarity Results

We additionally verify that our gradient-alignment results are not affected by KL regularization. Specifically, we compare the cosine between the minority-group gradient and (i) the KL-regularized PPO gradient in Figure 3, and (ii) the reward-only vanilla PPO gradient in Figure 5. The reward-only vanilla PPO gradient is

$$\nabla_\theta \left\{ \mathbb{E}_{x\sim\mathcal{D},\, y\sim\pi_\theta} \left[r_{\phi_g}(x, y) \right] \right\}$$

and the KL-regularized PPO gradient is

$$\nabla_\theta \mathcal{J}_{\text{RLHF}}(\theta) = \nabla_\theta \left\{ \left\{ \mathbb{E}_{x\sim\mathcal{D},\, y\sim\pi_\theta} \left[r_{\phi_g}(x, y) \right] - \beta\, \text{KL}(\pi_\theta \| \pi_{\text{ref}}) \right\} \right.$$

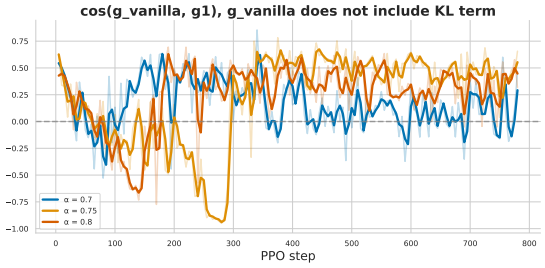


Figure 5: **Gradient alignment between vanilla objective and minority objective.** The plot reports the cosine similarity between the reward-only vanilla PPO gradient, not containing the KL regularization term, and the minority group reward gradient.

The two diagnostics show nearly identical trends over training, with matching rises, drops, and relative ordering across α. Although the scalar KL penalty can become comparable to the reward term late in training, including it does not qualitatively change the observed cosine dynamics. This suggests that KL regularization primarily affects the scale of the update rather than the qualitative direction of alignment in our setting.

C.2 Additional MaxMin-RLHF Training Results

Figure 6 and 7 show the group-level batch scores during MaxMin-RLHF training when $\alpha = 1.2$ and 1.3 respectively. Unlike vanilla RLHF optimizing a fixed pooled reward, MaxMin-RLHF updates

the policy according to the currently lower performance group. As a result, the minority and majority scores exhibit alternating improvements: when one group falls behind, subsequent updates tend to improve that group, and other group is ignored or even hurt. This switching behavior is especially clear when the two group scores cross or move in opposite directions during training. Overall, the plots support the interpretation that MaxMin-RLHF corrects worst-group performance through objective switching.

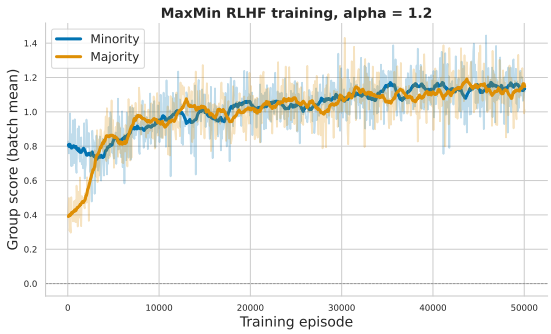


Figure 6: **Group scores during MaxMin-RLHF training,** $\alpha = 1.2$**.**

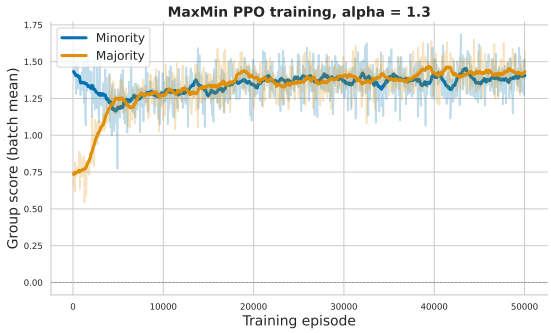


Figure 7: **Group scores during MaxMin-RLHF training,** $\alpha = 1.3$**.**

C.3 Additional Results under Preference Heterogeneity

We provide additional results comparing vanilla RLHF, MaxMin-RLHF and PCGrad-PPO across different values of the preference-divergence parameter α. Figure 8, 9 and 10 report overall reward, minority-group reward, and majority-group reward, respectively. Together, these results support the main-text conclusion that vanilla RLHF fails because the pooled objective becomes dominated by one group as preferences diverge, while group-aware or gradient-aware methods can partially correct this failure.

The minority-group plot shows the clearest collapse: vanilla PPO maintains high minority reward only when preferences are nearly identical, but its minority score drops sharply once α increases beyond the aligned regime. MaxMin-RLHF improves worst-group performance by shifting optimization toward the minority group, but it becomes unstable when preferences are negatively correlated. PCGrad-PPO provides the most stable minority performance, suggesting that directly modifying gradient directions is useful when group objectives interfere.

The majority-group plot shows that vanilla RLHF continues to achieve high majority reward even when minority performance collapses. This confirms that the failure is not a general optimization failure, but a group imbalance induced by the pooled RLHF objective. MaxMin-RLHF and PCGrad-PPO reduce this imbalance, with PCGrad-PPO preserving majority performance more consistently while also protecting minority reward.

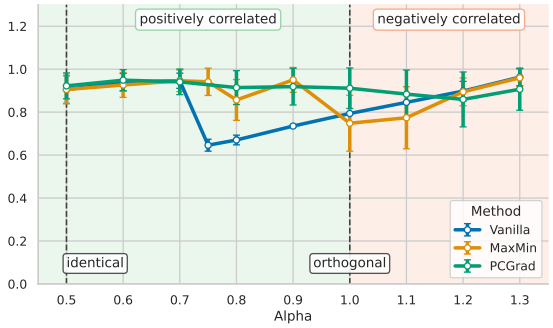


Figure 8: **Overall performance under preference heterogeneity.** Overall performance is computed as $0.8\times$ majority score + $0.2\times$ minority score, where 0.8 is the majority proportion. Vanilla RLHF drops in the intermediate divergence regime, while MaxMin-RLHF and PCGrad-PPO better preserve aggregate performance.

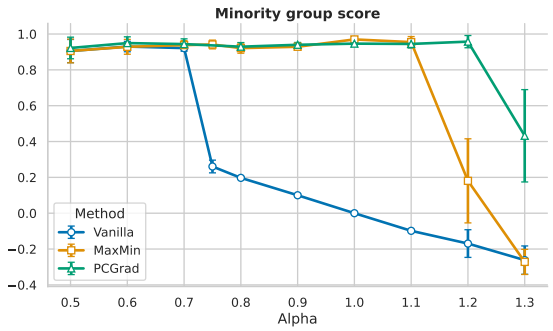


Figure 9: **Minority group performance under preference heterogeneity.** Vanilla RLHF exhibits a sharp minority-group collapse once preferences become sufficiently divergent, even before the negatively correlated regime. MaxMin-RLHF delays this collapse by emphasizing the worst-performing group, while PCGrad-PPO maintains the most stable minority performance across the tested range.

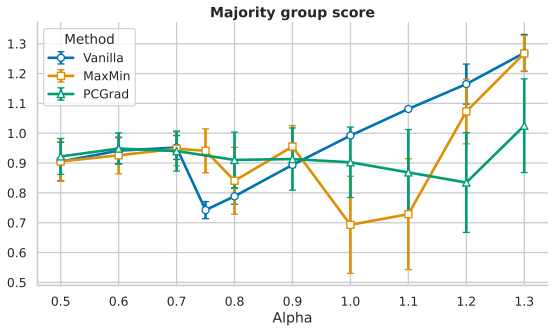


Figure 10: **Majority group performance under preference heterogeneity.** Vanilla RLHF continues to achieve high majority-group scores as preferences diverge. MaxMin-RLHF sacrifices majority reward when optimizing the worst-performing group at some points, whereas PCGrad-PPO maintains a more balanced majority performance.